AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956

SCHOOL OF ARTIFICIAL INTELLIGENCE COIMBATORE

Satellite-Driven Multi-Pollutant Forecasting for Industrial and Respiratory Disease Risk Mapping

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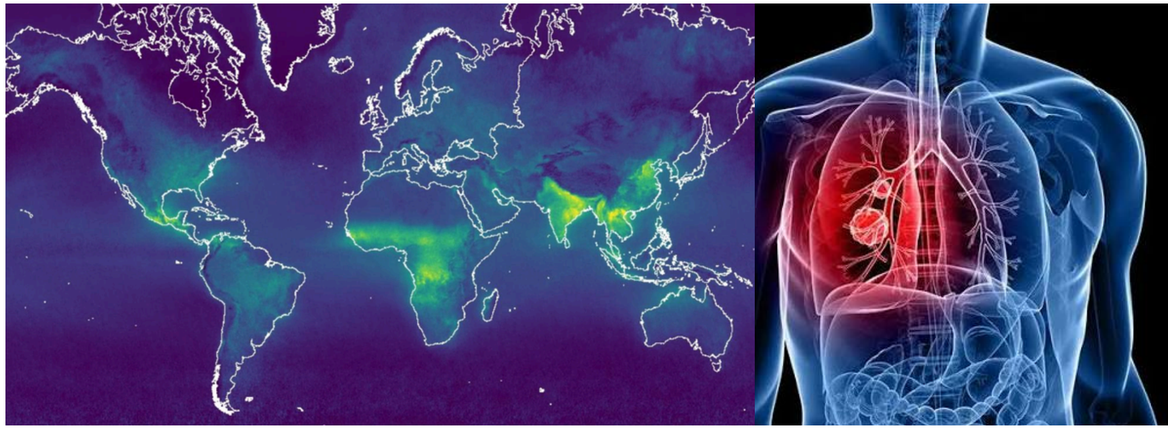
TEAM No. **B6**

Guided by Dr. Iswarya K. V.

Co-guided by Dr. Sajith Variyar V. V.

INTRODUCTION

- **Remote sensing satellites** estimate **atmospheric pollutant** concentrations over large spatial scales
- They measuring the interaction of **electromagnetic bands** with the Earth's atmosphere.
- **Sentinel-5P/TROPOMI and Sentinel-2** provide frequent observations of **NO_2, SO_2, and CO** overcoming the **limited coverage** of ground monitoring stations.



SIGNIFICANCE

Comprehensive Multi-Pollutant Coverage

Integrates **NO_2, SO_2, and CO** for a holistic assessment of air quality.

AI-Driven Forecasting

Utilizes **spatio-temporal deep learning** to predict future pollutant concentrations for proactive environmental monitoring.

Risk Assessment & Early Warning

Converts pollutant forecasts into **environmental risk levels and Early Warning Indicators (EWI)** for timely action.

Policy & Decision Support

Delivers **forecast-based insights** to support environmental policy, urban planning, and sustainable development initiatives(SDG Goals).

Risk Assessment & Early Warning to forecast multi-pollutant (NO_2, SO_2, CO) signatures from satellite imagery

MOTIVATION & PROBLEM STATEMENT

- **Sparse ground stations** limit localized pollution monitoring and forecasting.
- **Coarse Sentinel-5P observations** cannot capture fine-scale local pollution variations.
- **Pollution patterns vary by time horizon**, requiring different short- and long-term forecasting approaches.
- **Pollutants differ in data characteristics**, making a single forecasting strategy unsuitable for all.

Develop and evaluate a satellite-based, horizon-adaptive framework for multi-pollutant air-quality forecasting by combining Sentinel-5P atmospheric observations with Sentinel-2 spatial predictors

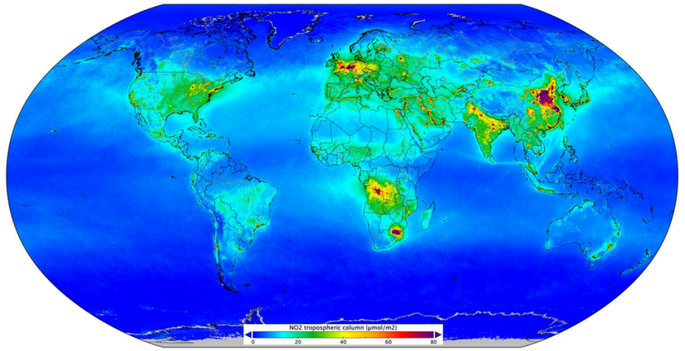
OBJECTIVES

Satellite Data Acquisition & Preprocessing
Acquire and preprocess Sentinel-5P pollutant observations and Sentinel-2 spatial predictors for Tamil Nadu.

Spatial & Temporal Analysis
Analyze pollutant distribution, missing data, temporal patterns, and spatial variability.

Multi-Horizon Forecasting
Develop short-term (1–5 days) and long-term (150 days) forecasting approaches for NO_2, CO, and SO_2.

Pollutant-Specific Model Evaluation
Compare forecasting approaches using R^2, Pearson correlation, SSIM, CSI, AUC and PR-AUC, as applicable.



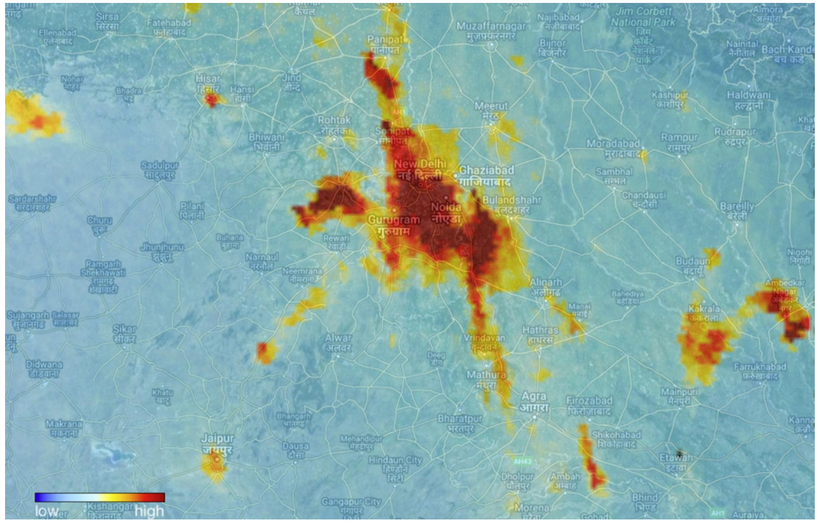
Global NO_2 distribution observed by the Sentinel-5P satellite.

LITERATURE REVIEW

Title	Year	Dataset(s)	Methodology	Limitation	Result
Nitrogen Dioxide (NO_2) Pollution Monitoring with Sentinel-5P Satellite Imagery over Europe during the Coronavirus Pandemic Outbreak	2020	Sentinel-5P TROPOMI Level-2 NO_2 (492 images), European ground monitoring stations	Sentinel-5P preprocessing, temporal NO_2 analysis, **GIS mapping**, validation with ground stations	Cloud cover, one observation/day, limited temporal resolution	Detected NO_2 reductions up to **85%** during COVID-19 and accurately mapped pollution hotspots.
Change Detection in Heterogeneous Optical and SAR Remote Sensing Images via Deep Homogeneous Feature Fusion	2020	Optical & SAR datasets from **Rikuzen-Takata**, **Iwate (Japan)**, and **Haiti earthquake**	**Deep Homogeneous Feature Fusion (DHFF)**, CNN-based feature extraction, IIST, One-Class SVM	Requires accurate image registration and high computational cost	Reduced false alarms and achieved superior Optical–SAR change detection over existing methods.
Satellite Monitoring for Air Quality and Health	2021	MODIS, VIIRS, MAIAC, OMI, TROPOMI, GOES-16, EPA AirNow, CTM models	Review of satellite, ground-monitor, and **model fusion** for $PM_{2.5}$ and NO_2 estimation	Cloud-induced missing data, indirect surface estimation, requires data fusion	Demonstrated that satellite data greatly improves large-scale air quality and health exposure assessment.
Change Detection within Remotely Sensed Satellite Image Time Series via Spectral Analysis	2020	Simulated vegetation data, MODIS NDVI, Landsat-8 EVI	Proposed **JUST** algorithm using **ALLSSA** for temporal change detection	Sensitive to window size; requires further validation	JUST outperformed BFAST by accurately detecting changes in irregular satellite time series without interpolation.

RESEARCH GAP

- Traditional studies rely on **sparse ground stations or coarse satellite observations**, limiting localized pollution analysis.
- Existing approaches focus mainly on **pollution monitoring**, with limited emphasis on **multi-horizon forecasting**.
- Different pollutants exhibit **different temporal and spatial characteristics**, making a single forecasting approach unsuitable for all pollutants.

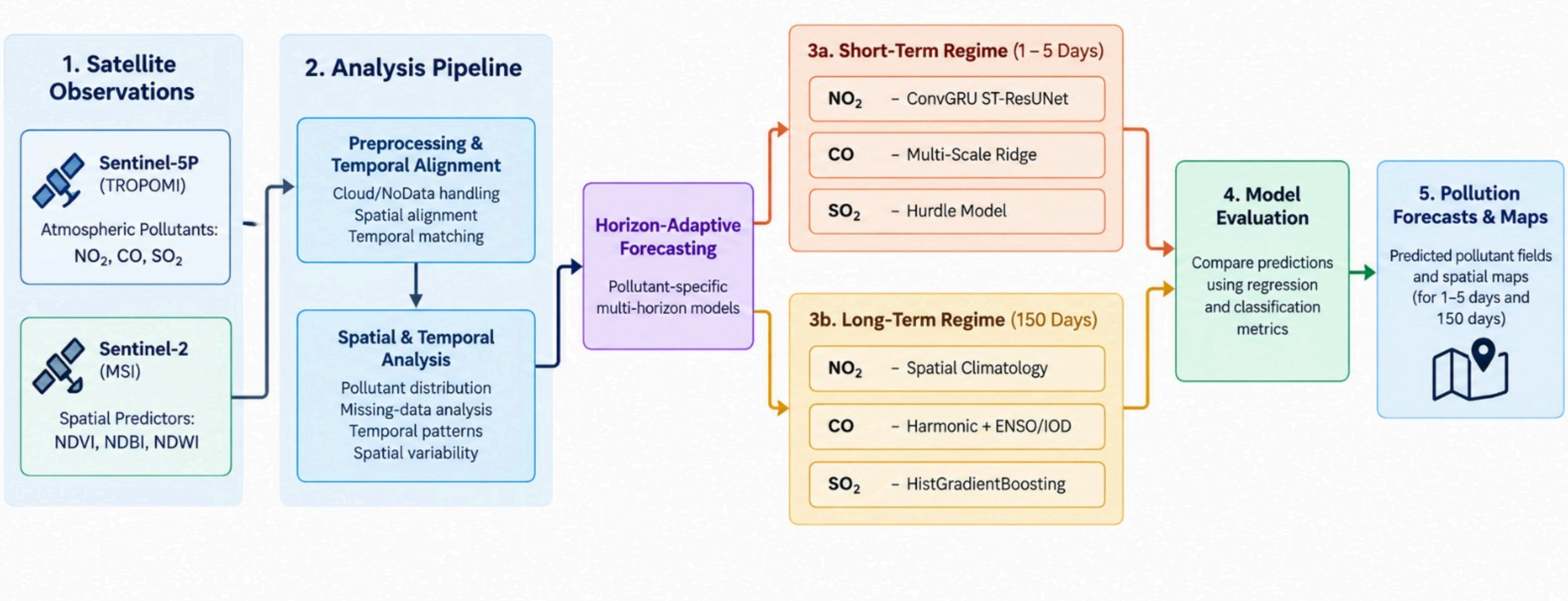


Pollution over Delhi from Sentinel-5P

<u>Research Opportunity</u>

- Develop a horizon-adaptive, pollutant-specific forecasting framework
- Integrate Sentinel-5P atmospheric observations with Sentinel-2 spatial predictors for improved pollution analysis and forecasting.

ARCHITECTURE DIAGRAM

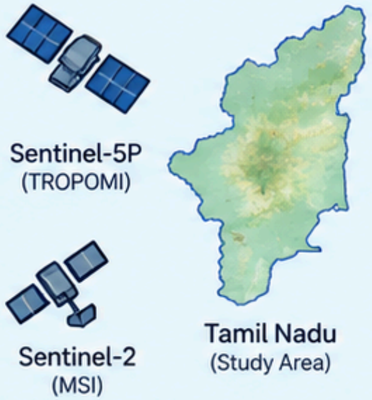


PROJECT PROGESS

1

DATA COLLECTION

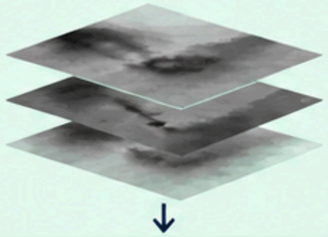
Collected Sentinel-5P and Sentinel-2 data for Tamil Nadu



2

DATA PREPROCESSING

Cloud/NoData handling, spatial alignment and temporal matching completed

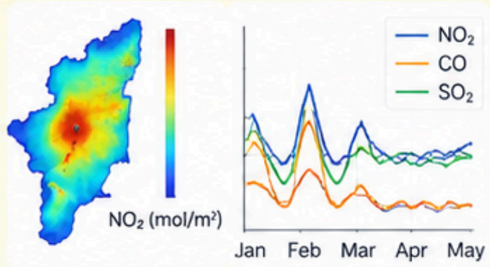


- Cloud masking
- Spatial alignment
- Temporal matching
- Feature extraction

3

SPATIAL & TEMPORAL ANALYSIS

Analyzed pollutant distributions, missing-data patterns, and temporal/ spatial characteristics



Spatial Distribution | Temporal Trends

4

FORECASTING MODELS

Developed short-term (1–5 days) and long-term (150 days) forecasting approaches for NO_2, CO and SO_2

Short-Term 1 – 5 Days — Initial Modelling

- NO_2 – ConvGRU ST-ResUNet
- CO – Multi-Scale Ridge
- SO_2 – Hurdle Model

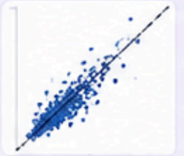
Long-Term Days — Ongoing

- NO_2 – Spatial Climatology
- CO – Harmonic + ENSO/IOD
- SO_2 – HistGradientBoosting

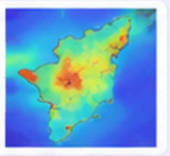
5

MODEL EVALUATION

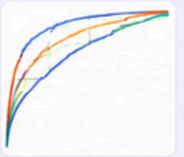
Evaluated forecasts using regression, spatial similarity and hazard-classification metrics



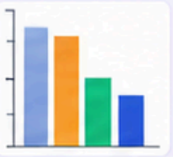
Regression Metrics



Spatial Similarity



AUC / PR-AUC

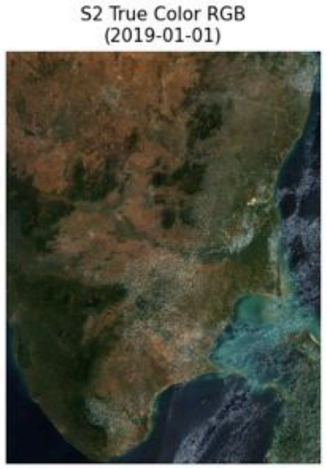


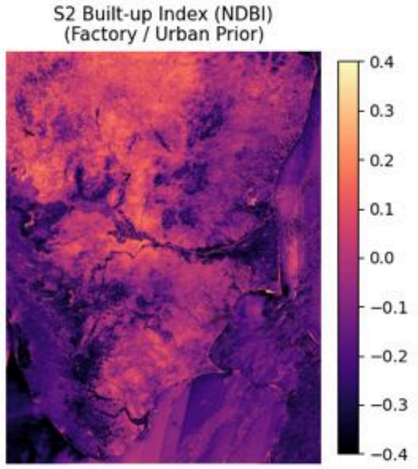
Comparative Analysis

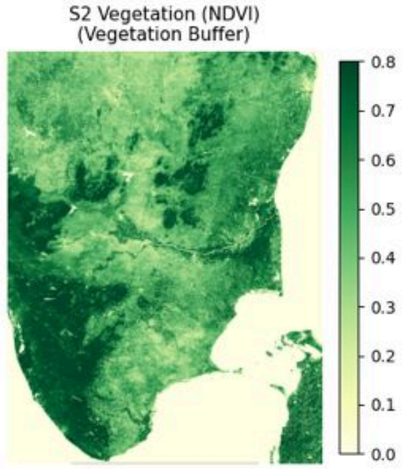
CURRENT STATUS | ✓ Data Collection → ✓ Preprocessing → ✓ Analysis → ✓ Model Development → ✓ Evaluation

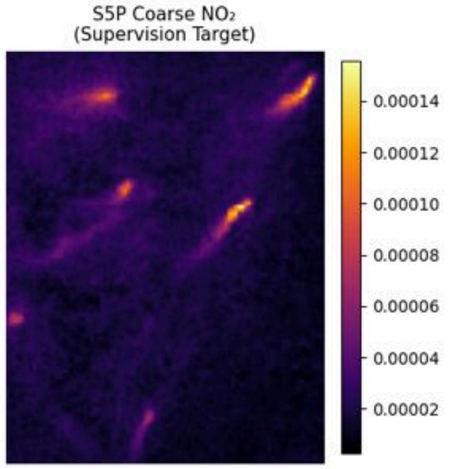
IMPLEMENTATION STATUS

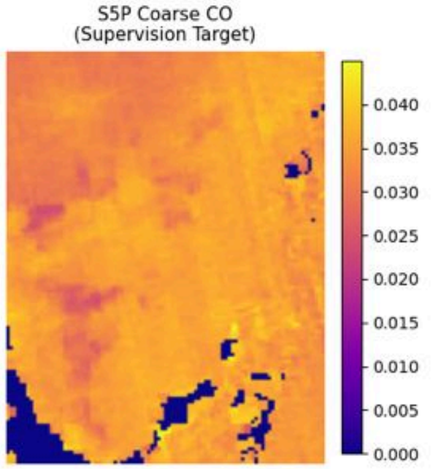
2019-01-01 to 2024-12-31









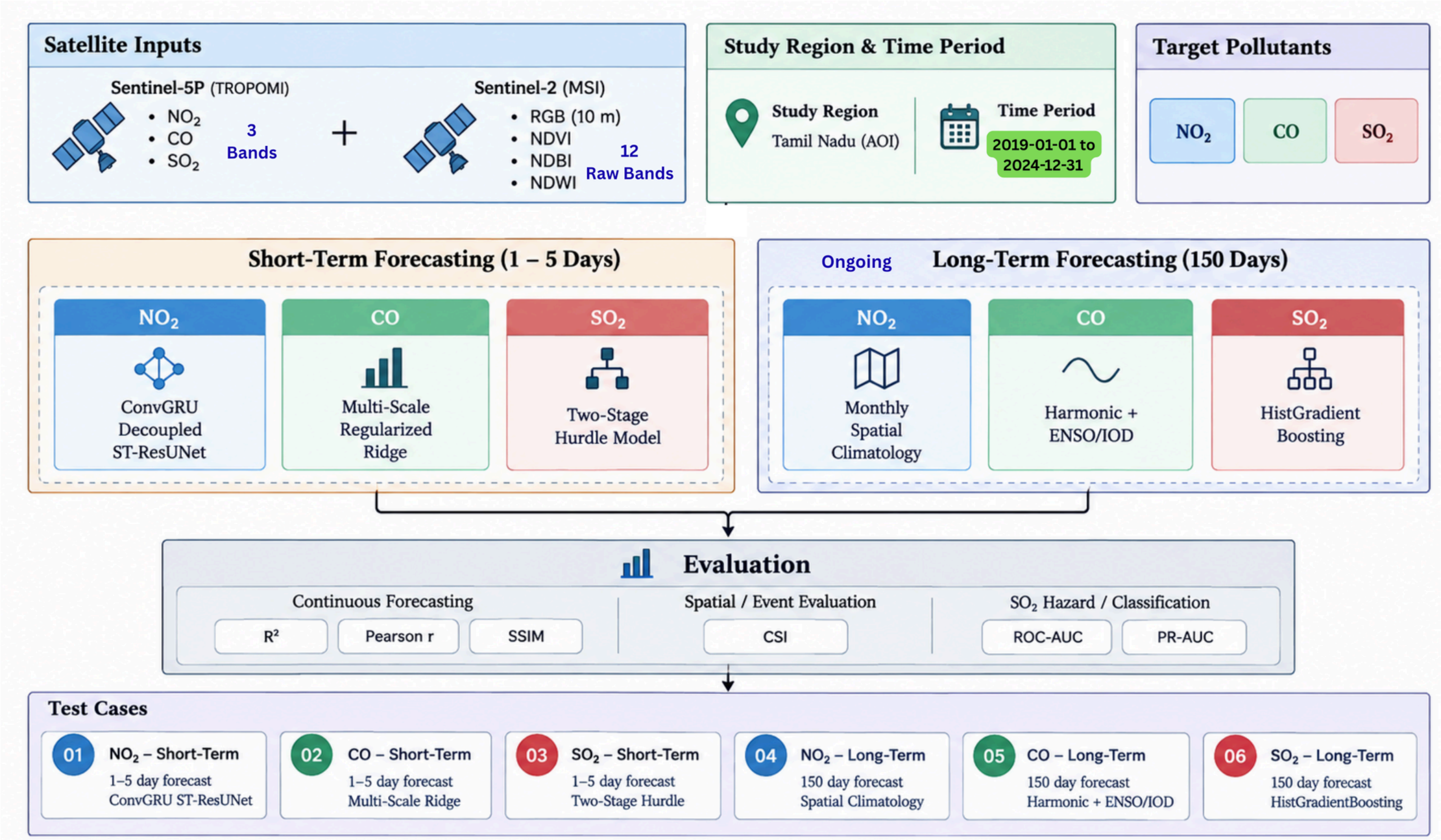


- TNAQ_S2
- TNAQ_S5P_CO
- TNAQ_S5P_NO2
- TNAQ_S5P_SO2

DATA COLLECTION

data/processed	feat: complete ST-ResUNet forecasting pipeline, multi-pat...	last week
docs	feat: complete ST-ResUNet forecasting pipeline, multi-pat...	last week
evaluation	Phase 4: Evaluate Attention Gates vs S2 Tap Depth; crown...	last week
plots	Phase 4: Evaluate Attention Gates vs S2 Tap Depth; crown...	last week
training	Phase 4: Evaluate Attention Gates vs S2 Tap Depth; crown...	last week
utils	feat: complete ST-ResUNet forecasting pipeline, multi-pat...	last week
.gitignore	Phase 4: Evaluate Attention Gates vs S2 Tap Depth; crown...	last week
README.md	Phase 4: Evaluate Attention Gates vs S2 Tap Depth; crown...	last week

EXPERIMENTAL SETUP



MODELS EXPLORED — SHORT-TERM (1–5 DAYS)

POLLUTANT	MODELS EXPLORED	APPROACH	CURRENT BEST MODEL
NO_2	• ConvLSTM • UNet • ST-ResUNet • Decoupled ST-ResUNet (ConvGRU)	Spatiotemporal deep learning to capture spatial patterns and short-term temporal dynamics	**ConvGRU Decoupled ST-ResUNet**
CO	• Linear Regression • Random Forest • XGBoost • Regularized Ridge (Multi-Scale)	Regularized regression with multi-scale spatial features to handle moderate sparsity and noise	**Multi-Scale Regularized Ridge**
SO_2	• Logistic Regression • Random Forest • XGBoost • Two-Stage Hurdle Model	Two-stage modeling for sparse observations: occurrence + concentration	**Two-Stage Hurdle Model**

Short-Term Selection: Models were selected based on each pollutant's spatial variability, temporal dynamics, and data sparsity.

MODELS EXPLORED — LONG-TERM (150 DAYS)

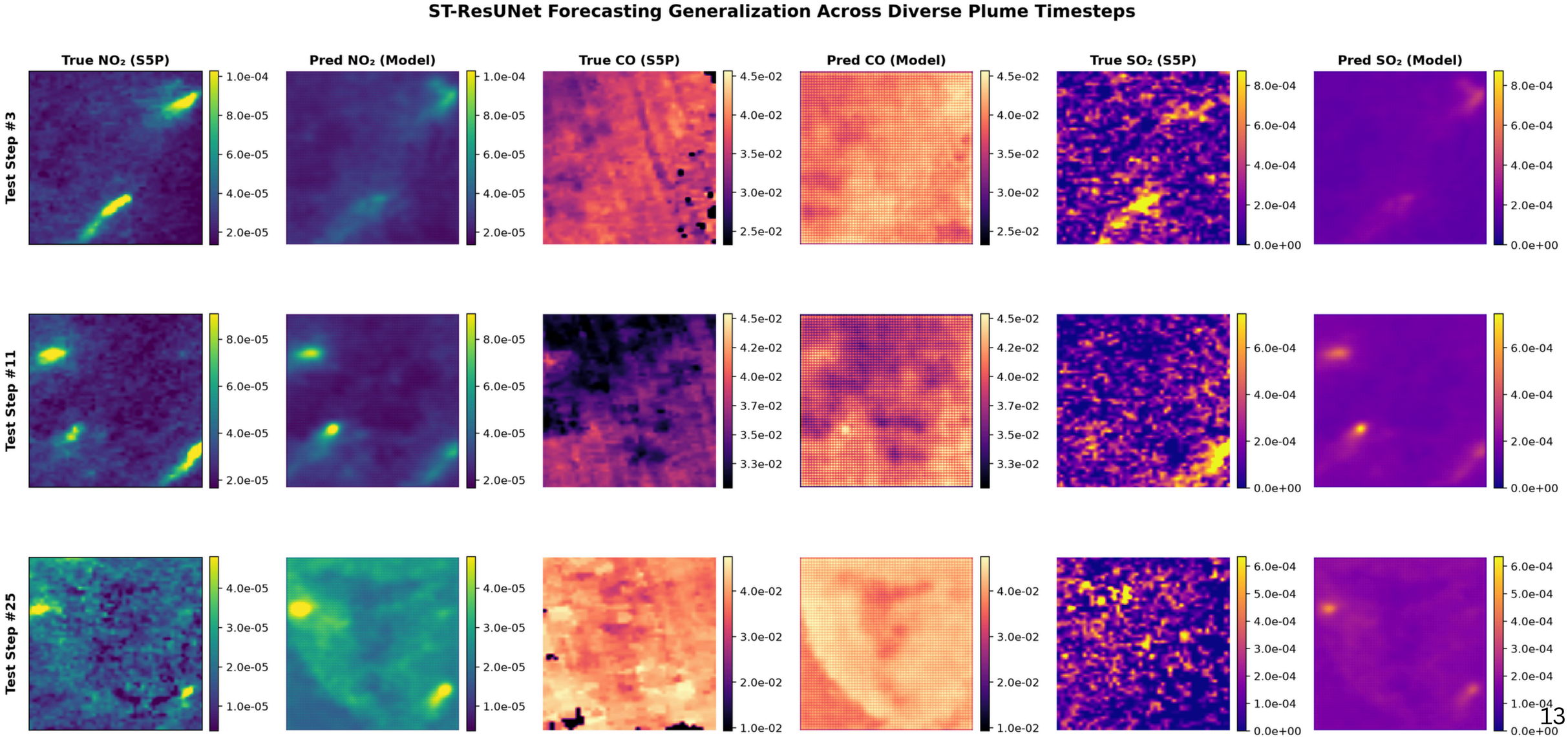
POLLUTANT	MODELS EXPLORED	APPROACH	SELECTED MODEL
NO_2	• Historical Mean (Baseline) • Seasonal Decomposition • Climatology (Monthly Spatial)	Monthly spatial climatology to capture seasonal and long-term spatial patterns	**Monthly Spatial Climatology**
CO	• Linear Trend • Seasonal/Harmonic Regression • Harmonic + ENSO/IOD • Gradient Boosting	Harmonic regression with climate indices (ENSO/IOD) to model long-term variability	**Harmonic + ENSO/IOD**
SO_2	• Logistic Regression • Random Forest • XGBoost • HistGradientBoosting	Gradient-boosted classification for long-term occurrence/hazard prediction	**HistGradientBoosting**

Long-Term Selection: Models were selected to capture seasonal, climatological and climate-driven patterns over the 150-day horizon.

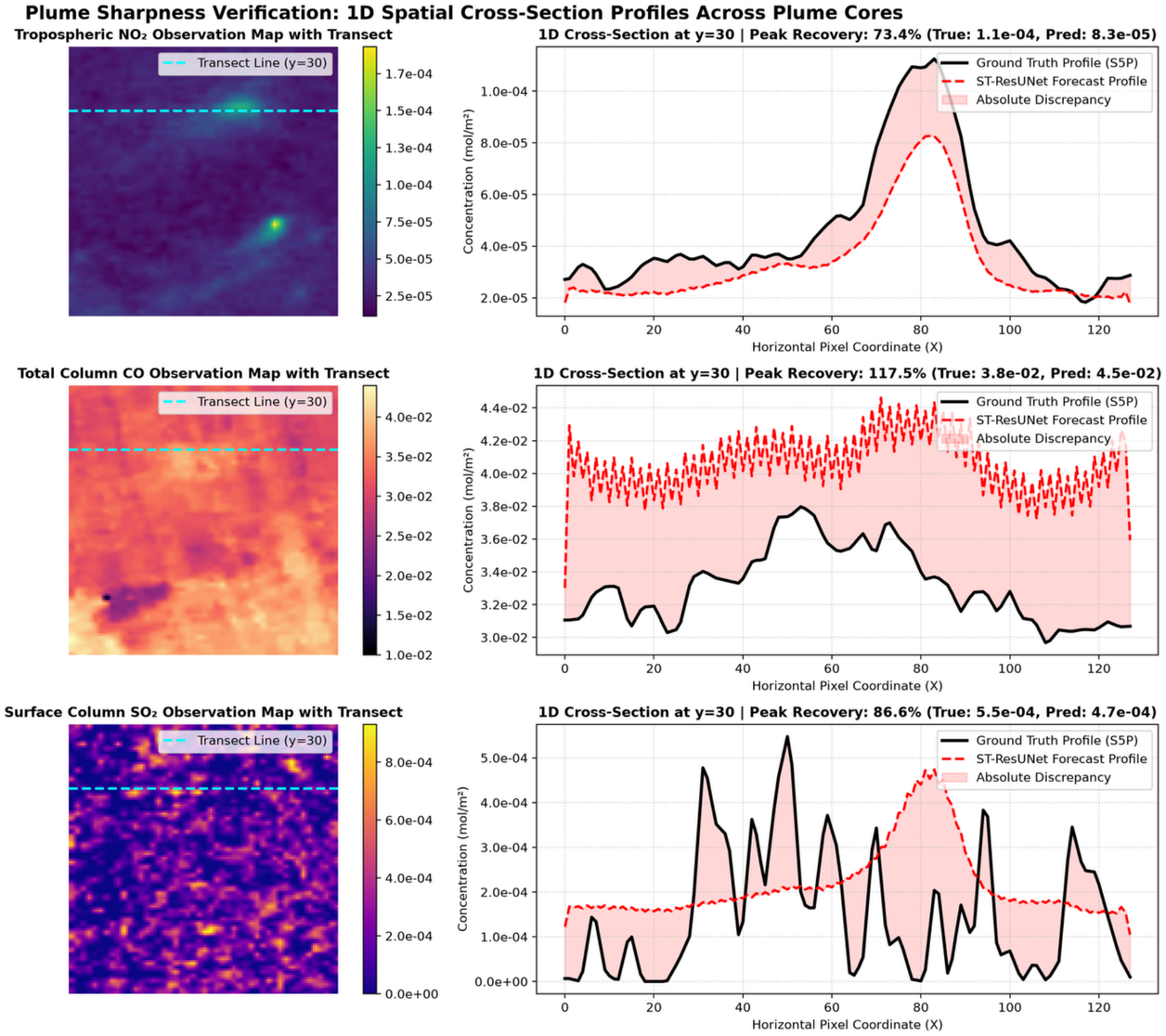
RESULTS – QUANTITATIVE PERFORMANCE

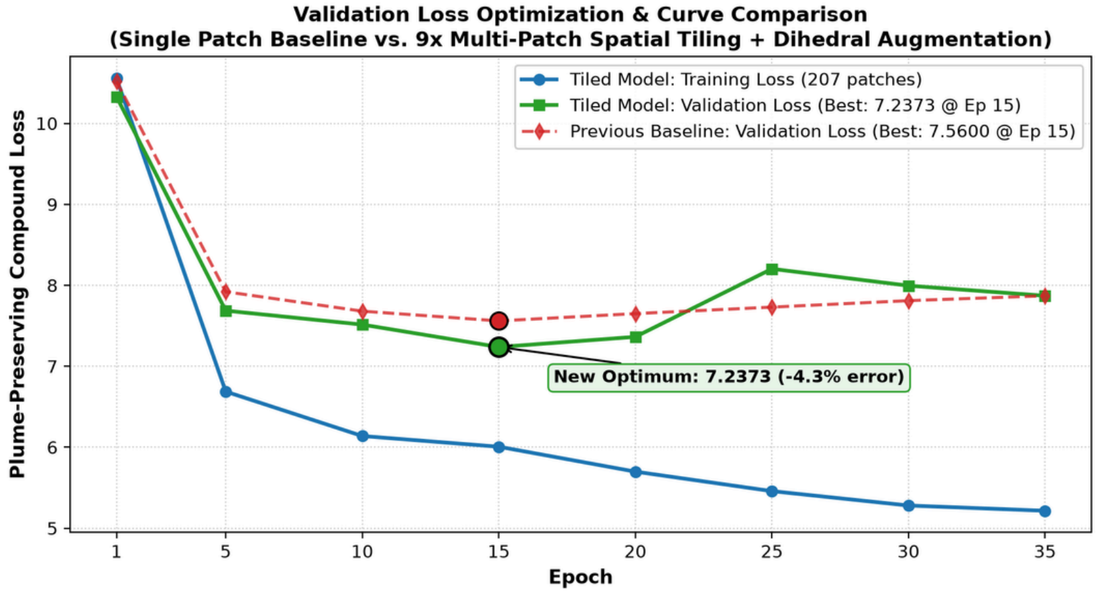
	POLLUTANT	MODEL	R^2	PEARSON R	SSIM	ADDITIONAL METRIC
short-term (1–5 days)	NO_2	**ConvGRU Decoupled ST-ResUNet**	0.651	0.810	0.896	CSI = 0.361
	CO	**Multi-Scale Regularized Ridge**	0.693	0.835	0.719	CSI = 0.091
	SO_2	**Two-Stage Hurdle Model**	N/A	0.275	0.256	AUC = 0.616
Long-term (150 days)	NO_2	**Monthly Spatial Climatology**	0.601	0.740	0.852	N/A
	CO	**Harmonic + ENSO/IOD**	0.683	0.836	0.711	N/A
	SO_2	**HistGradientBoosting**	N/A	N/A	N/A	ROC-AUC = 0.634; PR-AUC = 0.411; CSI = 0.261

RESULTS ANALYSIS & INTERPRETATION



RESULTS ANALYSIS & INTERPRETATION





CURRENT LIMITATIONS & TECHNICAL CHALLENGES

Challenge	Technical Impact
Missing Satellite Observations	Cloud cover and retrieval quality create invalid/NoData pixels, especially for SO_2.
Pollutant-Specific Data Characteristics	NO_2, CO and SO_2 have different spatial, temporal and sparsity characteristics, requiring different forecasting strategies.
Coarse Atmospheric Observations	Sentinel-5P provides coarse atmospheric measurements compared with Sentinel-2 spatial predictors.
Limited Fine-Resolution Ground Truth	Dense fine-resolution atmospheric ground truth is unavailable for direct validation of arbitrary high-resolution pollution predictions.

Key Challenge: Balancing satellite data quality, spatial resolution and forecasting horizon while maintaining reliable pollutant-specific evaluation.

REMAINING WORK & COMPLETION PLAN

MODEL REFINEMENT
Hyperparameter tuning
Model optimization
Improve forecasting stability

EXTENDED VALIDATION
Additional temporal testing
Spatial error analysis
Cross-pollutant comparison

ERROR & FAILURE ANALYSIS
Identify high-error regions
Analyze missing-data effects
Investigate SO_2 classification errors

FINAL OUTPUTS
Generate final forecast maps
Prepare comparative visualizations
Finalize prototype/demo

TEAM CONTRIBUTION & RESPONSIBILITY

Member / Role	AUGUST	SEPTEMBER	OCTOBER	NOVEMBER
Naresh Kumar V (Data & Pipeline)	Sentinel-5P & Sentinel-2 Data Collection	Data Preprocessing & Quality Control	Spatial Alignment & Temporal Matching	Dataset Preparation
Anto Rishath (Data Analysis)	Literature Survey	Exploratory Data Analysis	Temporal Pattern Analysis	Spatial Distribution Analysis
Antonio Roger (Model Development)	Forecasting Framework Design	Model Pipeline Development	Input Integration	Short-Term Model Implementation
Adarsh Pradeep (Research & Integration)	Literature & Research Analysis	Methodology Analysis	Pollutant Characteristics Analysis	Model Comparison

REFERENCE

[1] M. Vîrghileanu, I. Săvulescu, B.-A. Mihai, C. Nistor, and R. Dobre, "Nitrogen dioxide (NO2) pollution monitoring with Sentinel-5P satellite imagery over Europe during the coronavirus pandemic outbreak," Remote Sensing, vol. 12, no. 21, p. 3575, 2020, doi: 10.3390/rs12213575.

[2] X. Jiang, G. Li, Y. Liu, X.-P. Zhang, and Y. He, "Change detection in heterogeneous optical and SAR remote sensing images via deep homogeneous feature fusion," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 13, pp. 1551–1566, 2020.

[3] J. Evans, A. van Donkelaar, R. V. Martin, R. Burnett, D. G. Rainham, N. J. Birkett, and D. Krewski, "Estimates of global mortality attributable to particulate air pollution using satellite imagery," Environmental Research, vol. 120, pp. 33–42, 2013, doi: 10.1016/j.envres.2012.08.005.

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Thank You!